

TEJESWARISAI KOLLI

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer with 4 years of overall professional experience, including 4 years of expertise in Google Cloud Platform (GCP) Data Engineering and 2 years of hands-on experience in Generative AI. Proficient in developing enterprise AI solutions using LLMs, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), GPT-4o, ChromaDB, FastAPI, and Streamlit. Experienced in building multi-agent AI systems, AI-powered assistants, and intelligent data solutions while leveraging GCP technologies such as BigQuery, Dataflow, Cloud Composer, Cloud Spanner, Apache Beam, Python, SQL, and PySpark to deliver scalable end-to-end applications.

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Prodapt Solutions Pvt Ltd

Client: Verizon

Project: One Network Data

Jan 2026 – Present

- Engineered and managed Airflow DAGs in Cloud Composer to orchestrate Dataflow batch and streaming pipelines, reducing manual intervention across multiple GCP data domains.
- Implemented Apache Beam transforms to load and insert large-scale data into Cloud Spanner with high throughput, ensuring transactional consistency across distributed writes.
- Built a natural language query layer over BigQuery using LangChain and GPT-4o, enabling analysts to retrieve pipeline insights without writing SQL.
- Developed a RAG-based data lineage assistant with ChromaDB and prompt engineering to automate data quality reports and surface anomalies from Dataflow pipeline logs.
- Integrated Cloud Storage as a staging layer for intermediate transformations before downstream delivery to BigQuery and Cloud Spanner.

Project: Message Code Simplification

July 2025 – Dec 2025

- Architected an end-to-end Consumer Data Platform (CDP) on GCP, unifying customer data from EDW and CJCM golden sources into a single Customer Universe model.
- Built real-time ingestion pipelines using Apache Kafka and GCP Dataflow to process streams from CJCM, Cassandra, and POS systems at scale.
- Designed multi-layer curation logic to cleanse and validate raw customer data, improving downstream data quality and consistency.
- Created BigQuery views and aggregations powering Looker dashboards and ML models, enabling data-driven decision-making across teams.
- Enforced data governance by embedding validation rules at each pipeline stage, reducing data incidents and ensuring auditability.
- Activated unified customer profiles to Adobe Campaigns and Adobe Experience Platform, enabling personalised marketing at scale.

GenAI Training

May 2026 – Jun 2026

Capstone Project: SGEIA (Smart Grid Energy Intelligence Assistant)

- Completed Prodart's GenAI program covering Prompt Engineering, RAG, LangChain, LangGraph, multi-agent systems, Guardrails, DeepEval/Ragas evaluation, LangSmith/Langfuse observability, and FastAPI development.
- Built SGEIA using LangGraph StateGraph with GPT-4o, ChromaDB (hybrid BM25 + semantic RAG), XGBoost for grid stability prediction, and Isolation Forest for smart meter anomaly detection.
- Delivered a production-ready SGEIA system with FastAPI streaming backend and Streamlit dashboard, achieving >0.85 faithfulness score via DeepEval quality gates and full LangSmith observability across 5 agent nodes.

GCP Data Engineer

Proxima Systems Pvt Ltd

Client: SnippetSentry

Project: SnippetSentry*May 2022 – Dec 2024*

- Designed and delivered end-to-end data pipelines on GCP using Apache Airflow, BigQuery, and Cloud Storage, supporting high-frequency ingestion and analytics reporting.
- Managed Cloud Storage buckets, IAM roles, and access controls to maintain data security, compliance, and auditability across environments.
- Built and optimised 15+ BigQuery datasets, tables, and SQL queries, cutting average query execution time by 40% and improving analytics turnaround.
- Proactively monitored pipeline health, identified bottlenecks, and resolved failures — significantly reducing latency and improving data reliability.
- Authored runbooks, data flow diagrams, and test coverage for 10+ deployed pipelines, enabling faster onboarding and quicker incident resolution.
- Mentored new team members on Apache Airflow best practices, standardising development workflows and accelerating team productivity.

Project: AI Flow Mapper

Dec 2023 – Dec 2024

- Developed AI Flow Mapper, reducing manual diagram creation time by ~60% by automatically transforming requirement keywords and JSON inputs into structured mind maps and flow diagrams.
- Designed the data ingestion layer to parse and normalize JSON inputs, enabling dynamic generation of structured diagram outputs.
- Built backend GCP pipelines for transformation logic, storing intermediate results in Cloud Storage and serving outputs via BigQuery.
- Reduced requirements-to-diagram turnaround by 3x, enabling product and engineering teams to visualise complex workflows in minutes rather than hours.

Project: Bentley Systems

Feb 2022 – April 2022

Client: Bentley | Role: SQL Developer

- Wrote and optimized SQL queries to extract, manipulate, and analyse data from relational databases.
- Developed complex joins, subqueries, and aggregate functions to fulfil business reporting needs.
- Designed database schemas including tables, primary and foreign keys, and indexes.
- Performed data cleaning and validation on 5+ datasets, resolving inconsistencies and improving data accuracy for downstream reporting.
- Supported reporting dashboards by writing backend SQL queries for data sourcing and summaries.

TECHNICAL SKILLS

Cloud Platforms	Google Cloud Platform (BigQuery, Cloud Storage, Dataflow, DataProc, Pub/Sub, Cloud Spanner, Cloud Composer)
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Data Tools	Apache Airflow, Apache Beam, Apache Spark, Databricks, MongoDB
AI / GenAI	LangGraph, LangChain, OpenAI GPT-4o, Gemini, ChromaDB, RAG, Prompt Engineering, LangSmith, DeepEval, Ragas, Langfuse
AI Serving	FastAPI (SSE Streaming), Streamlit, uvicorn
ML & Evaluation	XGBoost, Isolation Forest, SHAP
BI & Automation	Power BI, Google Cloud Console, Looker
Programming	Python, SQL, JavaScript, PySpark
Others	Agile, GIT Labs, JIRA

CERTIFICATIONS

- Python (Core and Advanced) — National Skill Development Corporation
- Introduction to Artificial Intelligence — LinkedIn
- DataBricks Lakehouse - Coursera
- GenAI Program — Prodapt Solutions

EDUCATION

Bachelor of Technology – Electronics & Communication Engineering Andhra University, Visakhapatnam	2018 – 2022
Intermediate (MPC) Sri Chaitanya Junior College, Visakhapatnam	2016 – 2018